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1. Introduction

Nowadays, companies need reliable communication systems that allow their employees to share information and work together, even when they are located in different offices. A properly organized network and an appropriate application can help improve communication, reduce delays, and provide more organized access to company resources.

In this project, we propose designing a distributed enterprise platform that connects two company sites. The proposal combines a network infrastructure, an application developed in Go, and an authentication service built with FastAPI. These components will be planned to work together and provide a more organized and secure environment for communication between employees.

During this first phase, our group will focus on understanding the problem, identifying the main requirements, selecting the initial solution, assigning responsibilities, and organizing the work schedule. The technical implementation and detailed configuration will be completed progressively in the following phases.

This project will also help us apply the knowledge acquired during our studies and improve our skills in networking, programming, security, documentation, and teamwork.

2. Context and Problem Description

Companies with more than one office may experience communication and information-sharing difficulties when their network infrastructure and internal applications are not properly organized. Employees may have problems accessing authorized resources, communicating with colleagues from another site, or using services consistently.

For this reason, the company needs a solution that connects both locations and provides a clear way for employees to communicate. The network must organize devices according to their functions and allow communication between the sites under defined rules.

Another important aspect is security. Not every user should have unrestricted access to all company resources. The system should include authentication mechanisms to verify user identities and initial security measures to protect information.

Based on this situation, our group proposes planning a distributed platform that combines network connectivity, an internal communication application, and an authentication service. This proposal will establish the foundation for the development and evaluation activities planned for the next phases.

3. Justification

We consider this project important because communication is a basic requirement for companies that operate from different locations. A well-planned network can help employees access the services they need and communicate more efficiently.

The proposed internal chat application will provide a practical example of how software can support communication between employees. Developing the application in Go and integrating it with an authentication service built with FastAPI will also allow us to study how different technologies can work together.

From a technical perspective, the project will allow us to apply concepts such as IP addressing, network segmentation, routing, DHCP, NAT, firewall rules, and network security. It will also help us understand how to organize a software project and evaluate its operation.

Finally, we must consider the social and ethical aspects of the platform. Protecting user information, respecting privacy, and assigning access permissions appropriately are important responsibilities when developing a system for a business environment.

4. Objectives

4.1. General Objective

To design a proposal for a distributed enterprise platform that connects two company sites, integrates an application developed in Go with an authentication service built with FastAPI, and considers security measures, performance evaluation, and the social impact of the solution.

4.2. Specific Objectives

1. Identify the communication and information-access needs of the company at both sites.
2. Plan an organized network infrastructure that allows communication between the offices.
3. Propose a network and IP addressing structure according to the needs of each area.
4. Define how the enterprise application will integrate with the authentication service.
5. Establish initial measures to protect information and control user access.
6. Organize team responsibilities and establish a work schedule.
7. Define tests and metrics to evaluate the solution's performance in later phases.
8. Analyze the ethical and social responsibilities associated with using the platform.

5. Description of the Proposed Solution

As a group, we propose developing a platform composed of three main elements: a network connecting the two company sites, an enterprise application, and a service responsible for verifying user identities.

The network will allow devices at both offices to communicate according to established rules. The application will provide a way for employees to perform the activities defined for the project, while the authentication service will help verify who is attempting to access the system.

For the enterprise application, we propose an internal chat system for employees. This option addresses a specific need: improving communication between workers at the two sites. This is our initial proposal and will be confirmed before development begins.

The application will be developed in Go and integrated with an authentication service created with FastAPI. This approach will separate the main application functions from the process of verifying user identities (FastAPI, n.d.; The Go Authors, n.d.).

Regarding the infrastructure, we plan to use OpenWrt on Raspberry Pi devices or an equivalent authorized platform, according to the project requirements. The detailed network configuration will be developed during the following phases (OpenWrt Project, n.d.).

6. Project Scope

This section defines the elements included in the project and the activities that will remain outside the scope of this first phase.

6.1. Activities Included in the Project

The project includes:

- Planning a network infrastructure for two company sites.
- Organizing devices into separate networks according to their functions.
- Planning IP address allocation using the network block 10.10.0.0/16.
- Proposing communication between the sites using Open Shortest Path First (OSPF) routing.
- Planning network services such as Dynamic Host Configuration Protocol (DHCP), Network Address Translation (NAT), and firewall rules.
- Integrating an enterprise application developed in Go.
- Planning an authentication service using FastAPI.
- Defining connectivity, security, basic performance, and usability tests.
- Organizing team responsibilities, the project repository, and the documentation.
- Planning the statistical and sociotechnical analysis for later phases.

6.2. Activities Not Included in This First Phase

The purpose of this submission is not to present the platform as a fully implemented system. Instead, we will focus on the project context, the initial selection of the application, team organization, and planning.

Additional technologies such as load balancing, cloud infrastructure, Software-Defined Networking (SDN), and IPv6 will not be treated as main requirements because they are not necessary for the minimum scope defined for this project.

The detailed implementation, integration, configuration, and validation of the system will be completed progressively according to the course schedule.

7. Users and Stakeholders

To ensure that the proposed solution responds to the company's needs, we have initially identified the following stakeholders.

Stakeholder	Role in the project
Employees at the main site	Use the application and authorized company resources.
Employees at the second site	Communicate with colleagues and access permitted services.
Network administrator	Maintain connectivity and manage the network infrastructure.
Application administrator	Manage users and review the operation of the application service.
Development team	Design, program, document, and evaluate the solution.

Security personnel	Review access controls and information protection measures.

This identification is preliminary. During the requirements analysis, we will define each user's responsibilities and permissions more precisely.

8. Team Organization and Responsibilities

To develop the project in an organized way, we propose dividing responsibilities into different areas. Each team member will have primary tasks, but important decisions will be reviewed together to maintain a consistent solution.

Work area	Responsible member	Main activities
General coordination	José Vaca	Organize meetings, review progress, and coordinate submissions.
Networking and infrastructure	Pedro Cañar	Research and plan the network topology, addressing, and equipment.
Programming	José Vaca	Plan the Go application and define its main functions.
Security and authentication	Pedro Cañar	Research access controls and plan the FastAPI authentication service.
Statistics and social analysis	Dehivid Salgado	Propose metrics and analyze the project's social and ethical aspects.

Documentation and review	Entire team	Review the report, record decisions, and prepare the presentation.

Although responsibilities are divided, all team members will contribute to the project's progress. This will help us share knowledge, identify problems earlier, and complete the activities according to the agreed schedule.

9. Communication Plan

Communication is necessary to keep the team organized and avoid delays. For this reason, we will use shared digital tools to coordinate activities, exchange information, and review progress.

We propose the following communication rules:

- Use a shared messaging channel for quick questions and updates.
- Hold meetings when important decisions or technical problems need to be discussed.
- Record important decisions so that every member can review them.
- Assign tasks with a responsible person and an expected completion date.
- Inform the team when a task is delayed or when assistance is required.
- Review the project progress before submitting each deliverable.

If disagreements arise, we will discuss the different opinions and choose the most appropriate solution based on the project requirements. Our goal is to maintain respectful communication and ensure that all team members participate.

10. Initial Work Plan

The project will be developed progressively. During the first phase, the main priority is planning and organization rather than completing the technical implementation.

Activity	Expected result
Review the project requirements	Understanding of the technologies and activities required.
Define team responsibilities	A clear distribution of tasks among members.
Create the project repository	A shared space for organizing files and progress.
Establish communication agreements	Rules for meetings, messages, and conflict resolution.
Prepare the documentation	A report describing the context and initial planning.
Review the submission	Confirmation that the required elements are included.

The complete project is divided into different phases. Therefore, development, configuration, integration, and testing will take place progressively according to the schedule established by the course.

11. Required Resources

To work on the project, we will need technological resources, development tools, and methods for organizing the team.

The initial resources include:

- Hardware: Raspberry Pi devices or another authorized platform capable of supporting the required network configuration.
- Network software: OpenWrt and the tools needed to configure and test the network.
- Programming tools: Go and a development environment for creating the enterprise application.
- Authentication tools: Python and FastAPI for planning and developing the authentication service.
- Network testing tools: Utilities for checking connectivity, routing, IP addressing, and network services.
- Documentation tools: A shared repository and tools for preparing the report, diagrams, and project records.
- Team communication tools: Digital communication and collaboration platforms.

Before implementation begins, we will confirm which resources are available and which tools need to be installed or configured.

12. Initial Risks and Possible Solutions

During the project, several difficulties could affect the team's progress. Identifying these risks in advance will help us prepare possible solutions.

Risk	Possible impact	Preventive measure
Lack of coordination	Delays or duplicated tasks.	Assign responsibilities and establish deadlines.
Technical difficulties	Problems configuring or integrating components.	Research, test each stage, and document errors.
Unavailable equipment	Inability to practice certain configurations.	Confirm available resources and consult authorized alternatives.
Integration problems	The application and network may not work correctly together.	Define how the components will communicate before implementation.
File loss	Loss of the team's progress.	Use a shared repository and maintain backups.
Improper access to information	Exposure of data or unauthorized access.	Apply access controls and avoid storing unprotected passwords.
Disagreements among members	Delayed decisions and incomplete tasks.	Discuss issues respectfully, record agreements, and distribute work fairly.

These measures are preventive. Their effectiveness will be evaluated during the development and testing stages.

13. Academic Integrity and Responsible Use of Tools

As a group, we are committed to submitting work that we understand, can explain, and can review. Information obtained from external sources must be properly cited, and code or resources from third parties must be used according to their respective licenses.

If we use artificial intelligence tools to understand concepts, organize ideas, or review our writing, we will verify the results and follow the course instructions regarding their use and disclosure. We will not present invented results as if they were tests performed by our team.

We will also seek to protect user information, avoid using personal data without authorization, and apply appropriate measures to protect user accounts and the information managed by the platform.

14. Repository and Progress Tracking

To keep project files organized, we will use a shared Git repository. It will contain the documentation, diagrams, code developed during the following phases, and records of the tests performed.

The repository will be organized into folders with specific purposes, such as:

- app/ — the Go enterprise application.
- auth/ — the authentication service.
- network/ — network resources and configuration files.
- research/ — research and sociotechnical analysis.
- docs/ — reports, diagrams, and project documentation.

The repository must be created and configured as part of the team's activities. The final submission should include the actual repository URL and ensure that the instructor has the necessary access permissions.

15. Expected Results

At the end of the complete project, we expect to have a distributed enterprise platform that allows communication between the two company sites, a functional application, an integrated authentication service, and an organized network infrastructure.

We also expect to obtain data that will help us evaluate the system's operation and analyze its benefits, limitations, and possible effects on the people who use it.

For this first phase, the immediate expected results are:

- A clear definition of the problem.
- An initial selection of the enterprise application.
- Identification of the main users and stakeholders.
- Distribution of team responsibilities.
- Initial agreements for communication and collaboration.
- A preliminary work plan and identification of the required resources.
- Identification of potential risks and preventive measures.

These results will provide a foundation for the requirements analysis, design, development, integration, and validation phases.

16. Conclusion

As a group, we identified that the main need is to develop a technological solution that facilitates work between two company sites and allows employees to use the available services in an organized and secure way.

For this reason, we propose a platform that combines network infrastructure, an enterprise application developed in Go, and an authentication service built with FastAPI. We also consider it important to evaluate the system's operation and address aspects such as privacy, accessibility, and responsible information management.

Planning and distributing responsibilities will be essential for completing the project in an organized way. Although this first phase does not present the solution as already implemented, it defines the main elements that will guide the work in the following phases.

Finally, we believe this project will allow us to apply the knowledge acquired during our studies, strengthen our technical skills, and improve our ability to work as a team to solve a business need.

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